

Do Classical Price-Action Events Carry Directional Information?

A Pre-Registered Confluence Study on EUR/USD and GBP/USD

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Abstract

Discretionary traders often say that when two or three price-action patterns line up at the same level, the level matters more than when only one pattern fires. We pre-registered a study to test this claim head-on. The study screened seventy-six classical price-action events across four timeframes on EUR/USD between 2015 and 2021, then attempted to reproduce any survivor on a frozen 2022–2024 EUR/USD slice and on GBP/USD over the same screen window. An early version of the analysis returned forty-one survivors out of two hundred and eighty-four cells, including pairs of events that contradicted each other (every channel top and every channel bottom alive at the same time). That result is the visible footprint of a session-volatility confound: ATR-normalised forward movement at *random* fifteen-minute times varies by a factor of 3.7 across the 24-hour cycle. The protocol was amended to match each control draw to the event’s hour of day, and the survivor count dropped from forty-one to five. One of those five (`trendline_liquidity_sweep_low`) reproduced on the frozen EUR/USD slice; one (`channel_top_touch`) reproduced on GBP/USD. The effects are real but small (+0.05 to +0.35 units of average true range): useful as gate inputs to a separately validated trading agent, far too small to trade on alone. The repository never trades.

1 Introduction

This section explains in plain language what “confluence” means to a trader, states the research question, and previews the result.

What confluence means. A retail trader looking at a fifteen-minute EUR/USD chart will mark a few things on the screen: the top of an ascending channel, a 50 % fibonacci retracement line, the previous day’s high. When all three of those meet near the same price, the trader calls it a *confluence* and treats the level as high-confidence. The empirical content of “confluence” is therefore a comparative claim: a level with three classical patterns stacked should hold or reverse harder than a random level at the same time of day.

The research question. We asked two questions in nested form. First, does any individual classical price-action event carry directional information beyond a matched random baseline? This is **H1**. Second, do specific cross-timeframe combinations carry information their parts do not? This is **H2**, and is conditional on H1 producing survivors worth combining.

The headline result. Five of two hundred and eighty-four cells survived screening once the controls were corrected for time of day. One of those five reproduced on a frozen EUR/USD slice, one reproduced on GBP/USD, and all five kept the predicted sign on both out-of-sample looks.

The effects are too small to trade on standalone but are statistically real, and the survivors are concentrated in fifteen-minute trendline and channel geometry.

Outline. Section 2 states what was published before about price-action edge and multiple-hypothesis discipline. Section 3 defines the event dictionary, the outcome metric, the controls, and the statistical pipeline. Section 4 is a self-contained walk-through of the broken first analysis and the amendment that fixed it. Section 5 reports per-stage numbers. Section 7 lists what we cannot claim. Section 8 closes.

2 Background and prior work

This section sketches the published literature this study sits in, because most of what discretionary traders call “confluence” is not in any peer-reviewed paper.

The community vocabulary used here (zones, order blocks, fair-value gaps, market-structure breaks, liquidity sweeps, premium/discount, fibonacci retracements and extensions) is the Inner Circle Trader (ICT) glossary, taught publicly online for over a decade with no formal validation. Academic studies of FX price-action are sparser. Sweeney (1986) found simple filter rules in spot forex outperformed buy-and-hold net of costs over the 1970s. LeBaron (1999) extended that to moving- average rules with mixed results. Brock, Lakonishok and LeBaron (1992) applied a bootstrap test to technical trading rules on the Dow.

The multiple-hypothesis machinery used here comes from Benjamini and Hochberg (1995), whose false discovery rate procedure is the standard correction when many cells are screened at once. Bailey et al. (2014) catalogued how backtests overfit when this correction is skipped, and their framing of “minimum backtest length” motivated our split discipline. Lopez de Prado (2018) wrote down the broader research hygiene this lab tries to follow.

To our knowledge, no paper screens the full ICT-style event vocabulary under pre-registered multiplicity control and reports the survivor list honestly. This study is that screen.

3 Methodology

This section locks the event dictionary, the outcome metric, the controls, and the statistical pipeline. The full protocol was committed to the repository before any data was screened.

3.1 Event dictionary (Stage 0)

Eighteen detector modules emit seventy-six event types. The categories are: market-structure breaks and shifts, supply and demand zones, order blocks and breaker blocks, fair-value gaps, liquidity pools and sweeps, trendlines and parallel channels, horizontal support and resistance, chart patterns (triple tops, head and shoulders, triangles, wedges, flags, rectangles), fibonacci retracements and extensions, and candlestick patterns. Each event ships with a pre-registered directional hypothesis: “a touch should bounce”, “a break should continue”, “a sweep should reverse”, “a magnet should draw price”.

All detectors are causal: an event timestamped at bar t uses only bars $\leq t$. The dictionary was frozen before any Stage-1 statistic was computed. Fifty-one unit tests pin the detector contracts.

3.2 Outcome metric

For an event at bar t with directional hypothesis d , the maximum favourable excursion (MFE) is the largest price move in direction d over the next H bars, divided by the average true range (ATR) of the last 14 bars at t . Horizons by timeframe: $H = 30$ on D1, $H = 20$ on H4 and H1,

$H = 16$ on M15. A binary *hit* variable is also recorded: 1 if the event reaches +1 ATR before −1 ATR, 0 otherwise.

The ATR normalisation is what lets us compare across timeframes. It is also what made the first version of the analysis explode: see Section 4.

3.3 Controls

For each event a number of *control* draws is created from the same data. Each control is a randomly chosen bar with a direction sampled from the event cell’s own direction mix. The outcome metric is computed identically. The control is the null model: a level with no event attached.

The original protocol drew controls uniformly across all bars. We amended this to draw controls only from bars at the same *hour of day* as the event. The reason is in Section 4.

3.4 Statistical pipeline

A two-thousand-shuffle permutation test computes a one-sided p -value on the difference in mean MFE between events and controls within a cell. The Benjamini–Hochberg procedure controls the false discovery rate at 5 % across all adequately powered cells in the family, defined as cells with at least 100 events.

Verdicts are written into a four-tier registry:

- **alive:** positive effect, survived false discovery rate.
- **parked_weak_effect:** positive raw $p < 0.05$ but failed the corrected threshold. Recorded, not claimed.
- **parked_insufficient_n:** fewer than 100 events. Statistics recorded for future re-look.
- **dead:** negative or zero effect; closed.

3.5 Data splits

Table 1: Data splits used in the study. Each (pair, split) is consumed once.

Split	Range	Use
Screen	2015-01-01 – 2021-12-31	Stage 1 and Stage 2 selection
Confirm	2022-01-01 – 2024-12-31	frozen test of screen survivors only
Sealed	2025-01-01 onwards	untouched
Cross-pair	GBP/USD, same protocol	frozen replication arm

4 The session-volatility confound and amendment v2.1

This section is the worked example. It walks through what the original analysis returned, why it could not be true, and how the fix was specified before any new data was looked at.

4.1 The pathological first run

The first Stage 1 run produced 41 **alive** cells out of 284, an *a priori* implausible survivor rate. The survivor list included **channel_top_touch** and **channel_bottom_touch** both significantly positive on M15. A channel top is supposed to fade; a channel bottom is supposed to bounce.

Both cannot be alive at the same significance level on the same data unless the null is broken. The same pattern appeared for liquidity sweeps in both directions and for several pairs of fibonacci levels.

4.2 The diagnostic

A short diagnostic script measured the control MFE distribution on M15 hour-by-hour, ignoring event labels. The result was striking. At 19:00 UTC, average control MFE was 1.09 ATR. At 05:00 UTC, it was 4.03 ATR. That is a factor of 3.7 in the supposed null. The mechanism is mechanical: ATR(14) on M15 sums fifteen-minute ranges over the last 3.5 hours, so at the boundary between a quiet session and an active one, the next sixteen bars structurally exceed the ATR window. Events themselves are not uniformly distributed across the clock (the `asia_high_sweep` family, for instance, has 45 % of its events at 07:00 UTC), so any event family concentrated in active hours inherited a fake positive lift, and any family concentrated in quiet hours inherited a fake negative one.

4.3 Amendment v2.1

The protocol was amended once, documented in the repository before any re-analysis, to match each control draw to the event’s hour of day in addition to its direction. No detector parameters changed. No event outcomes were re-measured. Only the control-sampling code changed. The uniform-control run is preserved in `output/` as the cautionary record.

4.4 What the fix did

Table 2: Effect of the amendment on the survivor count.

Run	Alive cells		Notes
Uniform controls (original)	41 / 284		contradictory pairs alive
Hour-matched controls (amended)	5 / 284	all M15; trendline / channel / fib	

Five survivors is a believable number for a screen of 284 cells under 5 % false discovery rate. The contradictions disappeared.

5 Results

This section reports the numbers stage by stage, leading with the headline cell and then the table.

5.1 Stage 1 on EUR/USD screen (hour-matched controls)

Five cells alive, 16 parked weak, 77 parked for insufficient sample size, 186 dead. All five survivors are on M15:

5.2 Out-of-sample confirmation (EUR/USD 2022–2024)

`trendline_liquidity_sweep_low` confirmed at +0.308 ATR ($p = 0.0065$) on the frozen confirm window. The effect is essentially unchanged from the screen window (+0.303 in, +0.308 out), which is the signature of a stable effect rather than a lucky screen. The other four survivors shrank by roughly half out of sample, classic winner’s-curse attenuation.

Table 3: Stage 1 survivors on EUR/USD 2015–2021 under hour-matched controls. Effect is in units of ATR(14). All five passed the corrected threshold.

Event type	n	Effect (ATR)	p
<code>channel_top_touch</code>	19,793	+0.075	0.0005
<code>fib_50_tag</code>	5,134	+0.152	0.0005
<code>trendline_liquidity_sweep_low</code>	1,133	+0.303	0.0005
<code>fib_618_tag</code>	4,676	+0.124	0.0010
<code>fib_ext_1272_tag</code>	2,482	+0.189	0.0010

5.3 Cross-pair replication (GBP/USD 2015–2021)

`channel_top_touch` replicated on GBP/USD at +0.099 ATR ($p = 0.0005$). All five survivors kept a positive sign on this second out-of-sample look as well. Five out of five sign-consistency on two independent out-of-sample tests has probability roughly 3% under a random-sign null.

5.4 Stage 2 (cross-timeframe combinations)

The strict pre-registered Stage 2 was empty by construction. All five survivors are M15 cells, so there is no surviving higher-timeframe cell to use as context. An exploratory Stage 2 that included parked-weak H1 cells flagged `equal_highs_pool` as a candidate context for a future pre-registration but does not constitute a claim.

6 Discussion

The two validated cells are both reaction-to-geometry effects on M15. `trendline_liquidity_sweep_low` is a wick below an ascending support line that closes back above it; the rule predicts an extra roughly +0.3 ATR of upward excursion within four hours. `channel_top_touch` is the first touch of a projected parallel- channel upper boundary, and the rule predicts extra downward excursion of roughly +0.1 ATR on EUR/USD and roughly +0.1 ATR on GBP/USD.

Both numbers are small relative to spread. EUR/USD spread on M15 is roughly 0.1–0.2 ATR(M15) in active hours, which means a +0.1 ATR signal is mostly inside the cost band. These signals are inputs, not strategies. A live agent could use them as gate or exit inputs only after passing its own independent grid, holdout, and walk-forward chain.

The session-volatility result is itself a general methodological finding: intraday ATR-normalised metrics on M15 require time-of-day matched controls. Uniform-time controls quietly impose a false null.

7 Limitations

- Each detector is one auditable operational definition. A dead verdict closes that definition, not the underlying folk concept.
- Daily timeframe cells are structurally underpowered at $n \geq 100$ on seven years; a cross-pair panel design is the route forward for D1 claims.
- MFE measures opportunity, not net profit and loss; transaction costs are not modelled.
- Two looks at the screen window. The amendment was specified from a diagnostic on the screen window itself, so the post-amendment p -values are conditional on one analysis revision. The confirm and sealed windows were never used in the diagnostic.

- Indicator-only confluence and cross-family interactions are scheduled as future experiments (E008 and E009 in the repository registry) and not run here.

8 Conclusion

The headline is five cells out of two hundred and eighty-four, two of which independently reproduced under hour-matched controls. The discretionary claim that confluence “works” is partially supported on M15 trendline, channel, and fibonacci geometry. It is not broadly supported across the full ICT-style dictionary. Standalone M15 strategies on these signals are not viable after spread; informed gate or exit inputs to a separately validated trading agent are. The repository never trades and these findings will not be used live until they pass the agent’s own validation chain.

References

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